

Ayush Goel

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SUMMARY

Experienced Software Engineer (5+ years) specialized in designing and scaling large distributed systems, with expertise in AWS (EC2, EKS, S3, Lambda), microservices, container orchestration, and Infrastructure-as-Code. Proficient in modern architectural patterns (event-driven, serverless, microservices) and deeply knowledgeable about system reliability, performance tuning, and cost-optimization. Proven track record of leading cross-functional teams, delivering mission-critical solutions, and fostering a culture of high-quality engineering.

WORK EXPERIENCE

Software Development Engineer II

Amazon

US | Apr 2025 - Present

EC2 Team | Java, C++, Docker, Monitoring, Security

- Architected and launched Block Public Access security feature for AWS's newer networking fleets processing 100% of global AWS traffic, safeguarding millions of customer VPC resources from unintended internet exposure.
- Developed dataplane code and comprehensive testing framework (unit tests, integration tests, fuzzer tests), ensuring 99.99%+ uptime for infrastructure handling petabytes of daily traffic.
- Implemented observability infrastructure with metrics, real-time dashboards, and automated alarms, ensuring proactive detection and resolution of security policy violations across AWS's global fleet.

Senior Software Engineer

Goldman Sachs

Salt Lake City, US | Nov 2024 - Apr 2025

SDLC Team | Java, CI/CD, IaC, Kubernetes, AWS, Docker, Monitoring, Security

- Architected a high-throughput distributed artifact repository by redesigning our current artifact repository design for high-availability, achieving **99.99%** uptime and 10x faster dependency resolution for **100K+** daily build jobs.
- Designed a distributed pipeline lifecycle system using AWS Lambda, sharding GitLab's **1M+** daily pipelines by project for parallelized cleanup, reducing PostgreSQL storage costs by **50%** while maintaining **99.9%** API uptime during petabyte-scale deletions.
- Engineered a distributed pipeline query service bypassing GitLab API constraints with timestamp-based partitioning, achieving **15-second** latency (from **6 hours**) for **100K+** concurrent pipelines and ensuring sub-second SLAs for global trading platforms.

Software Engineering Intern

TDK InvenSense

California, US | May 2023 - Dec 2023

C++, Python, AWS IoT, ECS, Greengrass, S3, Lambda, CloudFormation

- Slashed operational costs by **30%** and tripled data ingestion rate by ingesting sensor readings with a data pipeline using AWS services, along with ROS2 C++ drivers.

Senior Software Engineer

Unicommerce eSolutions Pvt. Ltd.

India | Aug 2019 - Jul 2022

Backend Team | Early-Stage Tech Startup, E-Commerce, SaaS, Java, GraphQL, React

- Architected high-throughput, event-driven microservices using **AWS EventBridge** for routing, **SQS** for asynchronous messaging, and global load balancers for region-level fault tolerance, supporting **30%** increase in transaction throughput.
- Designed a **sharding** approach via **consistent hashing** to handle large data sets across SQL and MongoDB, achieving a **30%** improvement in query performance and data consistency.
- Augmented system efficiency by **50%** by redesigning **scalable, multi-threaded, distributed SaaS** backend microservices with **synchronous** and **asynchronous** integrations.
- Improved observability through a distributed logging strategy with **ELK Stack**, implementing tracing (OpenTelemetry) across microservices to reduce MTTR by **50%**.
- Achieved **30%** increased transaction throughput by leading **code** and **design** reviews and improvements as **Team Lead**.
- Attained **99.95%** uptime with robust monitoring and alerting via **Prometheus** and **Grafana**. Enhanced system observability and automated alerts, reducing mean time to detection by **50%**.
- Reduced on-call incidents by **15%** by creating comprehensive **Kibana** dashboards for **metric visualization**.
- Expanded customer base by **50%** by spearheading market expansion and partnerships with Amazon Marketplace, Reliance Industries, and Shopify, and leading cross-functional team collaborations.

TECHNICAL SKILLS

System Design & Architecture: Distributed Systems, Microservices Architecture, Event-Driven Design, CAP Theorem, High Availability, Fault Tolerance, CQRS, Sharding, Master-Slave Replication, Caching Strategies, Idempotent APIs, UML Diagrams

Programming: Java, Python, C++, GO, Shell Scripting, JavaScript, TypeScript

Backend Frameworks & Libraries: Spring, J2EE, Hibernate, JUnit, Mockito, GraphQL, REST, gRPC, MERN Stack

Infrastructure & DevOps: Docker, Kubernetes, EKS, Jenkins, CMake, Terraform, CloudFormation, Grafana, Prometheus, AWS IAM, ECR, CloudWatch, Valgrind, gdb

Data Storage & Processing: PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch, Kafka, Database Sharding, Change Data Capture

Cloud & Serverless: AWS (EC2, ECS, EKS, S3, RDS, Lambda, SQS, VPC, CloudWatch, X-Ray), GCP, Firebase, Connection Pooling

AI/ML: SageMaker, PyTorch, scikit-learn, LangChain, pandas, MLOps(MLflow, Kubeflow), Weights&Biases, Apache Airflow, Spark

Design Patterns: Authentication (OAuth2, JWT), Rate Limiting, Retry Mechanisms, Distributed Logging, Externalized Configuration

Version Control & Development: Git, Linux, Flutter, Android, CUDA

EDUCATION

University of Pennsylvania, United States | Master of Science in Robotics (Concentration in Computer Science & Artificial Intelligence)

Position: Teaching Assistant for Data Structures, Algorithms and Software Design; and Operating Systems